RoadPro

Road builder plugin for AutoCAD

User guide and command shortcuts

Version 3.0 · 07 October 2026

Type **RDP** to start



Developed by **www.exceltokml.com**

One command builds the complete road

Road and edges
Width both sides of the CL, CL linetype, surface hatch, chainage

Turning radius
Every bend filleted, default 10 m at the centre line

Barricades
Red and white units, black outline, left / right / median

Widening, tapers, junctions
Extra lane (RDW), width change (RDT), junction (RDJ)

Signs and markings
Sign posts and schedule, lane and edge lines, arrows

Radius and sections
C1 curve data and S1 straight data on the plan

Cross sections
Layers, crossfall, barricade and sign heights, dims

Report and Excel
Tables in the drawing plus a CSV file for Excel

Install and load in under a minute

1

APPLOAD

Type APPLOAD in the AutoCAD command line

2

Load

Select RoadPro.lsp and click Load

3

Startup

Contents > Add it to the Startup Suite to load in every session

4

RDP

Type RDP to open the RoadPro dialog

Before you start

- Full AutoCAD, Civil 3D or Map 3D 2013 or newer (AutoCAD LT is not supported)
- Type all sizes in metres. For millimetre drawings choose Millimeters in the dialog
- Save the DWG first: the Excel report is saved in the drawing folder

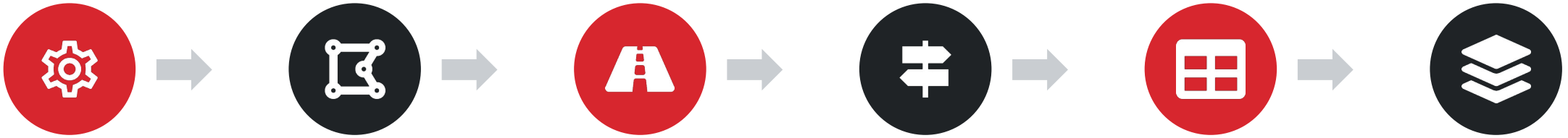
All commands and shortcuts

FULL COMMAND	SHORTCUT	WHAT IT DOES
ROADPRO	RDP	Build a complete road from a centre line
ROADREPORT	RDR	Report tables and Excel file for any RoadPro road
ROADRADIUS	RDRAD	Add radius, TC / CT and curve data to the plan
ROADSECTIONS	RDSEC	Add straight section labels S1, S2 … to the plan
ROADXSEC	RDX	Cross sections: typical, one chainage, every interval
ROADWIDEN	RDW	Lane widening with taper, fillet or direct link
ROADTAPER	RDT	Road taper: wider or narrower from any start to end
ROADJUNCTION	RDJ	Join a new road: junction / tie-in with kerb returns
ROADMARK	RDM	Road markings: edge lines, lane line, draw or remove
ROADPRORESET	-	Restore all settings to the factory defaults

Useful AutoCAD keys

U	Undo the whole road in one step
F2	Show the summary and file path
Enter	Finish the centre line or skip a prompt
Esc	Cancel the command

From centre line to finished drawing in six steps



1. Set options

RDP opens the dialog. Defaults are ready, press OK

2. Centre line

Pick points (Arc allowed, Enter to finish) or select a line

3. Auto build

Radius, edges, CL, surface, barricades, chainage

4. Signs and ends

Signs with IDs, tie-in or road closed, lane arrows

5. Report

Summary on screen, Excel file saved, place tables

6. Cross section

Pick a point to place the typical section

Tip: the whole sequence runs from one command. Steps 5 and 6 ask for a placement point; press Enter to skip the drawing tables or the section. The Excel file is saved either way.

Dialog settings: road geometry and barricades

Road Geometry

Road name / ID	**ROAD-01**
Road width	**7.30 m**
Turning radius	**10.0 m**
Radius measured at	**Centre line (CL)**
Drawing units	**Meters or Millimeters**
Centre line input	**Pick points or select**
Radius, TC / CT on plan	**On**
Straight sections on plan	**On**

Barricades

Sides	**Left + Right (Centre optional)**
Unit length	**2.00 m**
Unit width	**0.50 m**
Gap from road edge	**0.50 m**
Fill and outline	**Red / white, black**



Units alternate red and white. The count of each colour is in the report for ordering.

Dialog settings: signs, surface, chainage and outputs

Signs and Markings

Speed limit	**30 km/h**
Bend ahead	**30 m before**
Road works ahead	**On**
Chevron boards	**On**
Keep right (median)	**On**
Lane arrows	**On**
At tie-in	**STOP / GIVE WAY**
Symbol / spacing	**2.0 m / 15 m**

Surface and Chainage

Hatch road surface	**SOLID, 50%**
CL text in line	**On**
Chainage marks	**On**
Interval	**20 m**
Start chainage	**0 m**
Text height	**1.0 m**

Road Ends and Outputs

Start end	**Tie-in line**
Finish end	**Tie-in line**
Other options	**Closed / None**
Report + Excel	**On**
Typical section	**On**
Two-way traffic	**On**

Everything on its own layer

ROAD-CL
Centre line, CL linetype

ROAD-EDGE
Left and right edges

ROAD-SURFACE
Surface hatch

ROAD-BARRIER-RED
Red barricade fill

ROAD-BARRIER-WHITE
White barricade fill

ROAD-BARRIER-OUTLINE
Black outline

ROAD-CHAINAGE
Ticks and labels

ROAD-CURVE
Radius, TC / CT, C1

ROAD-SECTION
Straights S1, S2

ROAD-SIGN*
Posts, plates, symbols

ROAD-MARKING*
Lane arrows

ROAD-TIEIN
Tie-in line and E / N

ROAD-REPORT
Tables and legend

ROAD-XSEC*
Cross section parts

ROAD-WIDENING
Widening / taper labels

ROAD-JUNCTION
Junction labels, corner R

Signage placed automatically, traffic keeps right



Road works ahead

Entry of each direction



Speed limit

One spacing after entry



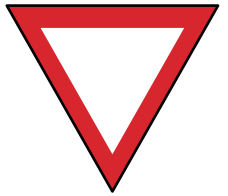
Bend ahead

Before each bend



STOP

Before each tie-in



Give way

Option at tie-ins



No entry

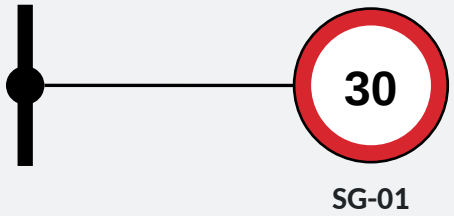
Far end, one-way road



Keep right

Median nose

How each sign is drawn on the plan



Post and plate
Black dot outside the barricade, plate faces the traffic

Leader, symbol and ID
IDs SG-01, SG-02 … follow the chainage order

Also drawn

- Chevron boards on the outside of bends
- ROAD CLOSED boards at closed ends
- White lane arrows on straights
- Sign schedule with E / N of every post

Radius and road sections on the plan

Every curve

RDRAD

```
C1
R inner = 10.000 m
R CL    = 13.650 m
R outer = 17.300 m
L (CL)  = 21.441 m
A       = 156.52 m²
Δ       = 90°00'00"
```

Radius line with arrow from the centre point, R value on the line, TC and CT ticks across the road with chainage.

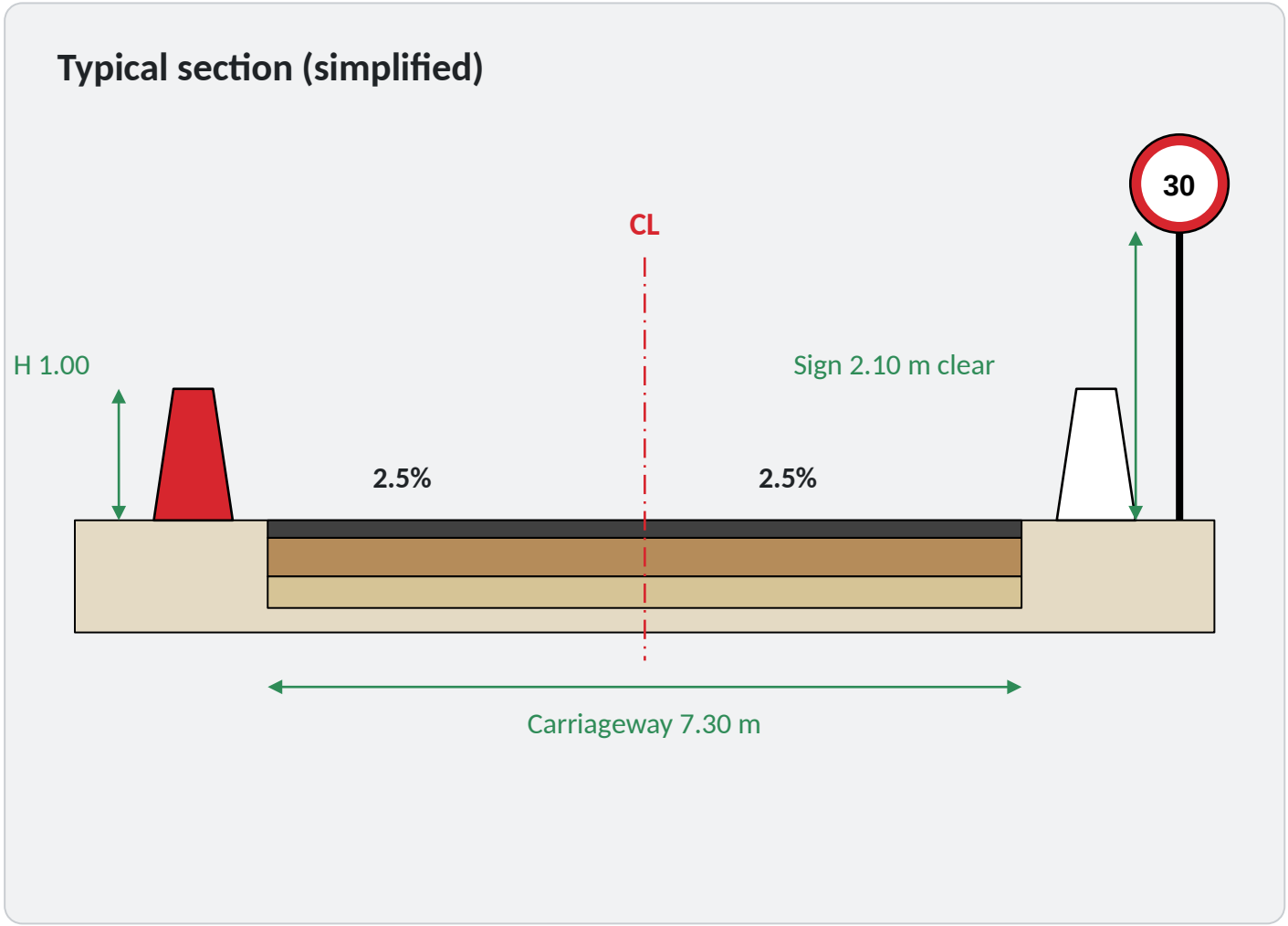
Every straight

RDSEC

```
S1  STRAIGHT
L = 45.230 m
A = 330.18 m²
CH 0+012.000 - 0+057.230
BRG 45°12'30"
```

Extent bar with end ticks on the right side, label parallel to the road. Bearing is whole-circle from North.

Cross sections with widths and heights



RDX **Three ways to draw**

1. **Typical**
   One generic section for the whole road

2. **One chainage**
   Type the chainage, note shows curve or straight

3. **Every interval**
   A sheet of sections, 3 per row, up to 60

Defaults: crossfall 2.5%, layers 50 / 200 / 150 mm, barricade 1.00 m high, sign clearance 2.10 m. Replace with your project values.

Lane widening with tapers and match points

Plan view (right side widening)



RDW **Select the centre line**

Side	**Left / Right / Both**
Extra width	**3.65 m**
Start / end chainage	**your input**
Taper in / out	**30 m / 30 m**
Taper type	**Straight or Curve**
Match point	**Fillet R or Direct link**
Fillet radius	**30 m**
Barricades	**moved to new edge**
Both sides	**own chainage + R**

Also drawn: new edge, lane surface, lane arrow, width dimension, match points and R=.. radius marks. Run RDW again for more sections (W1, W2 ...).

Straight + fillet

Arc with radius R at all 4 match points (TS, FS, FE, TE). Too big? RoadPro reduces R to fit.

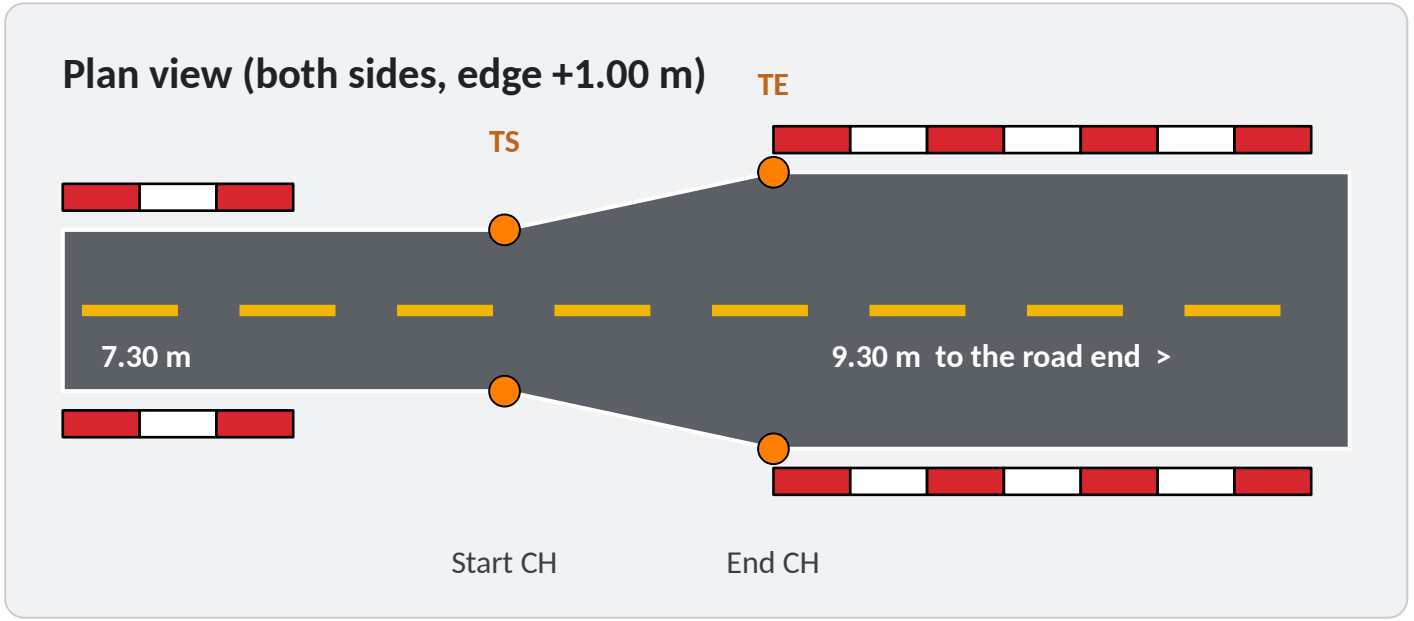
Straight + direct link

Sharp joints. The straight taper meets the edge directly at the match points.

Curve taper

Two reverse curves per taper, tangent at the match points, so no fillet is needed.

Road taper anywhere: wider or narrower



RDT **Select the centre line**

Side	**Left / Right / Both**
Taper start chainage	**your input**
Taper end chainage	**your input**
Edge shift per side	**+ wider / - narrower**
Taper type	**Straight or Curve**
Match point	**Fillet R or Direct link**
Fillet radius	**30 m**
Barricades	**follow new edge**
Both sides	**own chainage + R**

RDW = extra lane that comes back. RDT = width change that continues. Keep them on different chainage ranges of the same side.

Any start and end

Put the taper at any chainage: start, middle or end of the road.

Tapers add up

Run RDT again anywhere. 7.30 > 9.30 m, later 9.30 > 7.30 m.

Wider or narrower

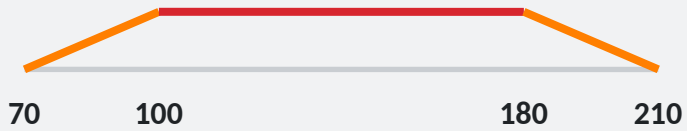
+ shift adds hatched surface, - shift wipes the old surface out.

Chainage input, both sides and radius marks

1. Type chainage your way

100	=	0+100
1250	=	1+250
1250.5	=	1+250.5

The taper is added outside your chainages:



Start 100, End 180, tapers 30 m. Keep Start ≥ taper in, and End + taper out ≤ road end.

2. Both sides, own values

Side = Both sides, then tick Different chainage on right side.

	LEFT	RIGHT
Start / End	100 / 180	150 / 260
Taper in / out	30 / 30	20 / 40
Fillet radius	30 m	15 m
Extra width	3.65 m	3.65 m

Works in RDW and RDT. Each side gets its own match points, labels and report rows (W1-L, W1-R).

3. Every radius marked



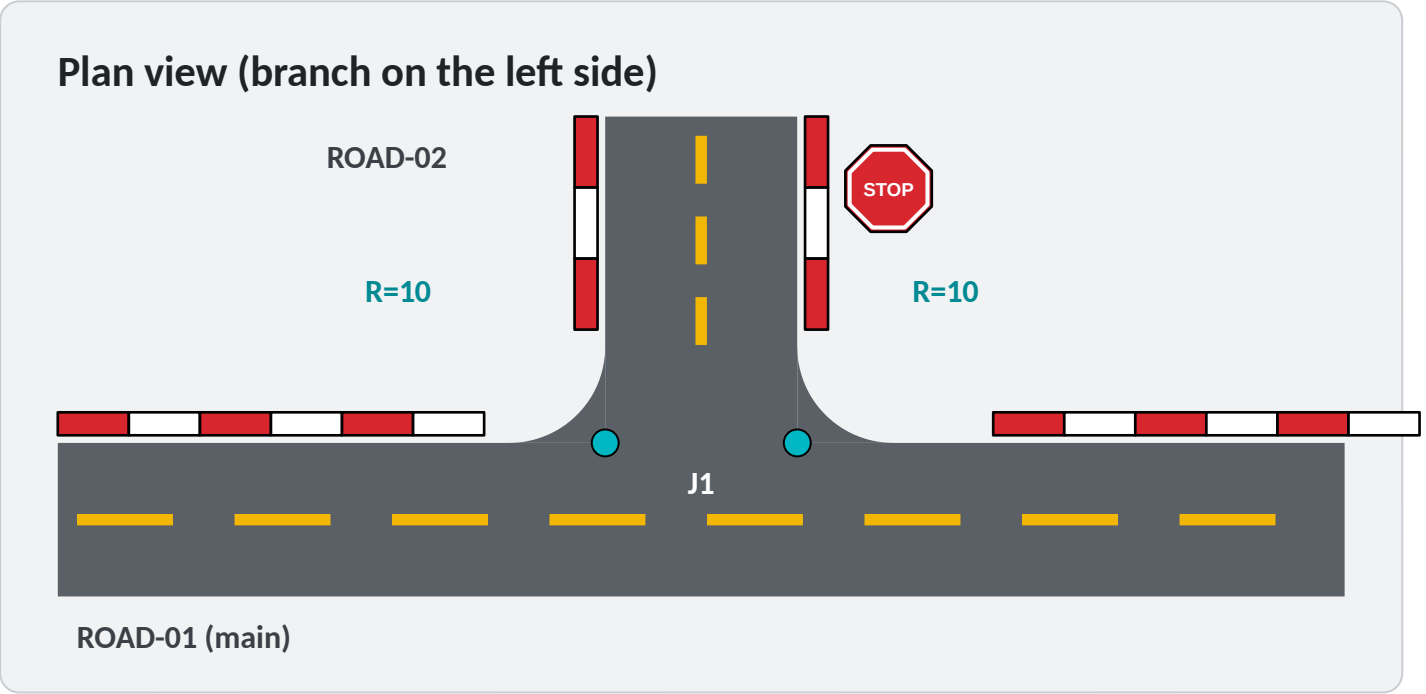
Straight + fillet: R at every fillet

Curve taper: R on each reverse-curve arc

Direct link: no radius, no mark

Arrow points to the new edge, text on the pavement side. If RoadPro had to reduce R, the mark shows the real R drawn.

Road junction / tie-in with kerb returns



RDJ Select the MAIN road

Branch name	**ROAD-02**
Branch width	**7.30 m**
Branch centre line	**Draw or select**
Corner (kerb) radius	**10 m**
Control on branch	**STOP / GIVE WAY**
Corner barricades	**On**
Branch chainage	**0+000 at junction**
Angle	**any (skew OK)**
Turning-area markings	**On**

Run RDJ again for more branches (J1, J2 …), on either side. For a crossroads add one branch on each side at the same chainage.

Start on the main CL

Draw the branch from the main road centre line outward, or select a line.

Everything automatic

Edge opened, kerb returns, corner surface, barricades and STOP / GIVE WAY.

Report and setting-out

TC, IP and CT of each corner with E / N, in Excel and in the drawing.

Road markings: lines, arrows and paint quantities

Along the road



Broken lane line 3 m / 6 m

Solid on bends and junction approaches

Turning area of a junction (RDJ)



Stop line + STOP (or give-way line + triangle), left / right turn arrow, broken edge line across the mouth.

RDM Draw or Remove

Line colour / width	**White, 150 mm**
Edge lines	**0.30 m inside edge**
Lane line	**Broken 3 m / 6 m**
Solid lane line	**Bends +/- 10 m**
Junction approach	**Solid 16 m**
With median	**No lane line**
Junctions	**Edge line gap**
Report	**Length + m2 paint**

New roads: tick Road markings in the RDP dialog. Existing roads: run RDM. RDJ redraws the markings of both roads with the junction gaps.

Report and Excel file

Summary

General, geometry, barricades, signs, road ends, chainage

Curve data

TC / CT chainage, radius, arc, deflection, E / N

Road sections

Every straight and curve: length, edges, area

Setting-out

E / N of CL, left and right edge at every chainage

Sign schedule

ID, sign, chainage, side, facing, post E / N

Widening, taper and junction setting-out

New edge offset, kerb returns TC / IP / CT with E / N

Road markings

Edge and lane line lengths, paint area in m2

Where is the Excel file?

`ROAD-01_RoadPro_Report.csv`

- Saved in the same folder as the DWG
- Enter only skips the drawing tables, never the Excel file
- Close the old CSV in Excel before running again
- Press F2 to see the full path

Run again any time: **RDR**

Quick fixes for common questions

Unknown command RDP

Load RoadPro.lsp with APPLOAD. Full AutoCAD only, not LT

Road is 1000 times too small or big

Choose Millimeters in Drawing units for mm drawings

Turning radius is too small

At the centre line, R must be more than half the road width

Black outline not visible

Dark background on screen. It still plots black

Excel file could not be saved

Close the old CSV in Excel, then run RDR again

Junction not formed

Start the branch ON the main road CL, away from widened or tapered edges

Widening runs outside the road

Start must be at least the taper in; End + taper out within the road

Chainage not accepted

Type 100 or 0+100. Older versions take plain metres only

Quick reference card

RDP	**RDR**	**RDRAD**	**RDSEC**
Build road	Report + Excel	Radius labels	Straight labels
RDX	**RDW**	**RDT**	**RDJ**
Cross sections	Lane widening	Road taper	Junction / tie-in
RDM	**U**	**F2**	
Road markings	Undo whole road	Summary + path	



Default values

Road width	**7.30 m**
Turning radius	**10 m at CL**
Barricade unit	**2.0 x 0.5 m**
Chainage	**every 20 m**
Speed limit	**30 km/h**
Crossfall	**2.5%**
Widening / taper	**3.65 m / 30 m**
Reset all	**ROADPRORESET**

Type RDP, accept the defaults, draw the centre line. RoadPro does the rest.

Developed by https://www.exceltokml.com/